Mechanical-Hydraulic Modeling of an Embankment Mower in Simscape

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***Abstract*—Accurate simulation of heavy duty hydraulic manipulators is often hindered by the difficulty of capturing the nonlinear interactions between hydraulic and mechanical dynamics. This paper focuses on the physical modeling of the hydraulic embankment mower, aiming to create a simulation environment that behaves like the real machine. By grounding the mechanical subsystem in CAD-derived geometry, we ensure that inertial properties and joint constraints reflect the actual hardware. The hydraulic domain moves beyond simple kinematic assumptions to capture dominant physical phenomena, including variable bulk modulus, line compressibility, and the finite response time of valve spools. The result is a tightly coupled plant where mechanical loads and hydraulic pressure transients drive one another. To verify the model's fidelity a standard strict closed-loop controller is applied to the system. The resulting emergent behaviors, such as hydraulic system delays, confirm that the model provides a valid, rigorous foundation for future digital twin applications.**

***Index Terms*—hydraulic manipulators, bulk modulus, closed-loop controller, hydraulic system delays, digital twin.**

I. Introduction

Embankment mowers represent a distinct category of hydraulic manipulators designed for operation within unpredictable settings. These systems, in contrast to industrial robots operating in regulated environments, encounter stochastic disturbances and uncertainties stemming from irregular terrain, fluctuating vegetation density, and varying impact loads [1]. Although hydraulic actuation is the prevailing method for such demanding applications, owing to its favorable power-to-weight ratio and inherent durability [2], this presents considerable dynamic complexities. The system's performance is dictated by significant nonlinearities, encompassing fluid compressibility, valve metering hysteresis, and temperature-dependent viscosity [3], [4]. Inaddition, the interaction between the hydraulic drive and the mechanical structure is bidirectional meaning the actuator forces drive the kinematic chain, while the inertial and cutting loads induce pressure transients that fundamentally alter the fluid properties. An accurate physical modeling of these coupled domains is a prerequisite for safety verification, performance optimization, and the development of autonomous functions [5].

In recent research, multi-domain simulation environments like Simscape and Modelica have become the preferred tools for virtual prototyping, replacing complex mathematical methods. Platforms allow for the simultaneous solving of equations related to rigid multibody mechanics and hydraulic networks, which helps maintain energy conservation across different physical areas [6]. Previous research has extensively documented the successful use of these combined models in earthmoving machinery, particularly excavators and forestry cranes [7]–[9].Embankment mowers possess distinct kinematic architectures and operational profiles that are characterized by long-reach cantilevered configurations and continuous-duty slewing are underrepresented in existing research. The coupling between high-frequency cutting vibrations and the hydraulic compliance seen in these machines is still not well understood.

The objective of this work is to bridge this gap by developing a high-fidelity, mechanically and hydraulically consistent model of the Mulag FME500 (hydraulic embankment mower). The modeling approach used in this study is fundamentally physical, as opposed to phenomenological. The mechanical subsystem is modeled using CAD-derived geometry, by ensuring that inertial characteristics and joint limitations approximately represent the physical hardware [10]. Concurrently, the hydraulic subsystem uses simplified linear models, explicitly accounting for significant nonlinearities, like variations in effective bulk modulus, line compressibility, and valve spool dynamics. Therefore, a highly integrated plant is achieved, wherein mechanical loads and hydraulic pressure fluctuations mutually influence each other, thereby mirroring the "stiff" yet oscillatory behavior characteristic of actual hydraulic machinery.

To validate the physical integrity of the proposed model, a baseline joint-level controller is implemented. It is important to note that the contribution of this study lies in the modeling methodology rather than the control strategy itself. The closed-loop controller serves as a validation instrument, verifying that the virtual plant exhibits realistic emergent behaviors such as delays and tracking errors of the end effector trajectory under varying bulk modulus and mass of the mower head.

II. Mechanical Modeling and Multibody Dynamics

The Mulag FME500 is modeled as a 5-DOF serial link manipulator. To bridge the gap between mechanical design and control validation, this study employs an automated model generation pipeline. The mechanism is decomposed into five rigid bodies: the front mounting frame (base frame), the swing joint, boom I, boom II, and the cutter head, corresponding to link 0 through link 5, respectively for simplicity.

Unlike simplified lumped-mass approximations, this work utilizes a CAD-to-simulation workflow. Geometric and inertial properties are extracted directly from the design data using the Blender/Phobos add-on [11]. This approach ensures that the simulation accurately reflects the asymmetric mass distribution and cross-coupling inertia terms ($I_{xy}, I_{yz}, \dots$) inherent in heavy agricultural manipulators.

A. Rigid Body Dynamics and Equations of Motion

The mechanical system is formulated using the Newton–Euler recursive algorithm within the Simscape Multibody environment. The dynamics of the manipulator in joint space $\mathbf{q} \in \mathbb{R}^5$ are governed by:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \boldsymbol{\tau}_{\text{fric}}(\dot{\mathbf{q}}) = \boldsymbol{\tau}_{\text{act}} - \mathbf{J}^T(\mathbf{q})\mathbf{F}_{\text{ext}}, \tag{1}$$

where:

- $\mathbf{M}(\mathbf{q})$ is the symmetric, positive-definite inertia matrix derived from the imported CAD tensors,
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ represents the Coriolis and centrifugal effects,
- $\mathbf{g}(\mathbf{q})$ is the gravitational torque vector,
- $\mathbf{J}(\mathbf{q})$ is the geometric Jacobian mapping task-space forces to joint torques.

A critical distinction in this modeling approach is the source of the actuation torque $\boldsymbol{\tau}_{\text{act}}$. Rather than assuming ideal torque generators, $\boldsymbol{\tau}_{\text{act}}$ is dynamically coupled to the hydraulic states derived in Section III:

$$\tau_{\text{act},i} = f_{\text{geom}}(F_{h,i}, \mathbf{q}), \tag{2}$$

where $F_{h,i}$ is the axial force generated by the i-th hydraulic cylinder and $f_{\text{geom}}(\cdot)$ represents the nonlinear mapping from linear cylinder force to revolute joint torque via the mechanism's lever arms.

B. Hydraulic Cylinder Kinematics

Each double-acting hydraulic cylinder is modeled explicitly as a kinematic a closed-loop chain connecting two adjacent links. The closed-loop convention is spherical joint-prismatic joint-cylindrical joint. This convention is specifically used due to the strictness of position constraints in Simscape. The barrel base is attached to link k via a spherical joint, the piston rod is attached to link $k{+}1$ via a cylindrical joint, and the relative motion along the cylinder axis is represented by an intermediate prismatic joint:

$$\text{link } k \;-\; S_{\text{base}} \;-\; P_{\text{cyl}} \;-\; C_{\text{rod}} \;-\; \text{link } k{+}1.$$

The attachment frames of S_{base} and C_{rod} are exported from the CAD model (via Blender/Phobos) and imported into Simscape as rigidly attached markers. This construction allows the cylinder to self-align under load while avoiding over-constrained kinematic loops, which is critical for numerical robustness in closed-chain agricultural mechanisms.

Let $\mathbf{p}_A(\mathbf{q})$ and $\mathbf{p}_B(\mathbf{q}) \in \mathbb{R}^3$ denote the positions of the barrel base and rod eye, respectively, expressed in the inertial frame. The instantaneous cylinder length is given by

$$\ell_{\text{cyl}}(\mathbf{q}) = \big\|\mathbf{p}_B(\mathbf{q}) - \mathbf{p}_A(\mathbf{q})\big\|, \tag{3}$$

and the prismatic joint coordinate s is defined as the relative extension with respect to a reference length ℓ_0 (e.g., fully retracted configuration),

$$s(\mathbf{q}) = \ell_{\text{cyl}}(\mathbf{q}) - \ell_0. \tag{4}$$

The unit vector along the cylinder axis is

$$\mathbf{e}_{\text{c}}(\mathbf{q}) = \frac{\mathbf{p}_B(\mathbf{q}) - \mathbf{p}_A(\mathbf{q})}{\ell_{\text{cyl}}(\mathbf{q})}. \tag{5}$$

Let $\mathbf{J}_A(\mathbf{q})$ and $\mathbf{J}_B(\mathbf{q})$ denote the translational Jacobians of points A and B. The scalar kinematic Jacobian that maps joint velocities to cylinder extension rate is

$$J_{\text{cyl}}(\mathbf{q}) = \mathbf{e}_{\text{c}}^{\mathsf{T}}\big(\mathbf{J}_B(\mathbf{q}) - \mathbf{J}_A(\mathbf{q})\big), \tag{6}$$

such that $\dot{s} = J_{\text{cyl}}(\mathbf{q})\,\dot{\mathbf{q}}$.

The axial force F_h generated by the hydraulic submodel acts along $\mathbf{e}_{\text{c}}$. Its contribution to the generalized joint torques is obtained from the virtual-work principle as

$$\boldsymbol{\tau}_{\text{act}} = J_{\text{cyl}}^{\mathsf{T}}(\mathbf{q})\,F_h, \tag{7}$$

which is the explicit form of the geometric mapping $f_{\text{geom}}(\cdot)$ in (2). In the Simscape implementation, the R–P–R structure is instantiated using two revolute joints and a prismatic joint oriented along the CAD-derived cylinder axis, and the reaction force measured at the prismatic joint is fed back to the hydraulic pressure network.

C. Cylinder-Induced Stiffness and Solver Configuration

The axial stiffness of a double-acting hydraulic cylinder arises from both the mechanical structure and the compressibility of the trapped fluid. Around a given operating point $(\mathbf{q}^\star, s^\star)$, the cylinder is linearized as

$$F_h \approx F_h^\star + k_{\text{cyl}}(s - s^\star), \tag{8}$$

where k_{cyl} is the small-signal axial stiffness. For a cylinder with effective chamber areas A_A and A_B, chamber volumes V_A and V_B, and hydraulic bulk modulus β, a commonly used small-signal stiffness model is [2]

$$k_{\text{cyl}} \approx \beta\left(\frac{A_A^2}{V_A} + \frac{A_B^2}{V_B}\right). \tag{9}$$

In this work, k_{cyl} is not taken from textbook beam formulas but is obtained in a simulation-consistent manner. At representative boom configurations, a small virtual displacement Δs (on the order of a few millimeters) is imposed on the prismatic coordinate while keeping the joint angles fixed, and the resulting change in axial force ΔF_h from the coupled hydraulic model is evaluated. The stiffness is then estimated numerically as

$$k_{\text{cyl}} \approx \frac{\Delta F_h}{\Delta s}, \tag{10}$$

which automatically accounts for the current pressure state, dead volumes, and any nonlinearities in the hydraulic submodel.

The equivalent joint-space stiffness contribution of a cylinder is obtained by combining (7) and (8), yielding

$$\mathbf{K}_q(\mathbf{q}) = \frac{\partial \boldsymbol{\tau}_{\text{act}}}{\partial \mathbf{q}} \approx J_{\text{cyl}}^{\mathsf{T}}(\mathbf{q})\,k_{\text{cyl}}\,J_{\text{cyl}}(\mathbf{q}), \tag{11}$$

so that the linearized joint-space dynamics around $\mathbf{q}^\star$ can be written as

$$\mathbf{M}(\mathbf{q}^\star)\,\delta\ddot{\mathbf{q}} + \mathbf{D}(\mathbf{q}^\star)\,\delta\dot{\mathbf{q}} + \big(\mathbf{K}_q(\mathbf{q}^\star) + \mathbf{K}_g(\mathbf{q}^\star)\big)\delta\mathbf{q} = \delta\boldsymbol{\tau}, \tag{12}$$

where $\mathbf{D}$ collects damping terms and $\mathbf{K}_g$ denotes the gravitational stiffness.

The eigenvalues of the generalized stiffness matrix $\mathbf{M}^{-1}\mathbf{K}_q$ define the fastest mechanical time scales of the coupled hydraulic–mechanical system. Due to the combination of large cylinder stiffness and relatively slow boom motion, this spectrum is typically wide, leading to stiff dynamics. Following common practice for hydraulic manipulators [7], a variable-step, implicit solver based on backward differentiation formulas is therefore employed in Simscape. This choice allows the simulation to resolve the fast cylinder-compliance dynamics without excessively restricting the global step size imposed by the slower rigid-body motion, while remaining consistent with the DAEs generated by the R–P–R cylinder representation.

D. Joint Friction and Interaction Dynamics

Accurate friction modeling is essential for realistic simulation, particularly for the swing joint which experiences high side-loads. The friction torque $\boldsymbol{\tau}_{\text{fric}}$ incorporates viscous damping and Coulomb effects:

$$\tau_{f,i} = \sigma_{v,i}\dot{q}_i + \tau_{c,i}\tanh(\gamma\dot{q}_i), \tag{13}$$

where the hyperbolic tangent function is used to approximate the signum function, ensuring solver stability at zero-velocity crossings [12].

The cutter head interaction with the vegetation is modeled as a variable external wrench $\mathbf{F}_{\text{ext}}$. While previous works often model this as a constant load, this study implements a state-dependent resistance model to simulate the stochastic nature of cutting dense vegetation:

$$\mathbf{F}_{\text{ext}}(t) = \mathbf{F}_{\text{nom}} + \boldsymbol{\delta}(v_{\text{feed}}, \rho_{\text{veg}}), \tag{14}$$

where $\boldsymbol{\delta}$ represents disturbance forces proportional to the feed velocity v_{feed} and vegetation density ρ_{veg} [1].

E. Kinematic Configuration

The manipulator topology follows a standard open-chain configuration. The joint coordinate frames are aligned such that the z-axis represents the axis of rotation, consistent with the Denavit–Hartenberg (DH) convention. The kinematic limits are strictly enforced using hard-stop constraints in the multibody solver to replicate the physical cylinder stroke limits. Figure 1 shows the fully assembled mechanical model of the Mulag FME500 Embankment Mower.

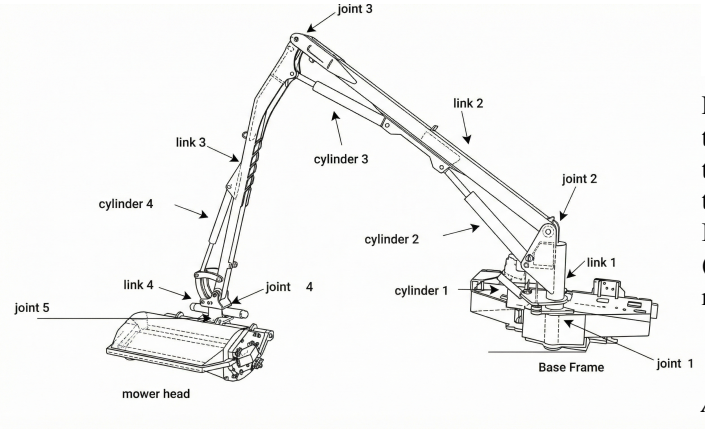


Fig. 1: Kinematic topology of the Mulag FME 500. The mechanism is defined as a serial chain comprising five revolute joints (Joint 1–5).

III. Hydraulic System Modeling and Dynamic Formulation

This section details the mathematical modeling of the hydraulic actuation system employed in the Mulag FME500 robotic manipulator. Unlike simplified kinematic models often used in mobile robotics, this study proposes a high-fidelity dynamic formulation that couples nonlinear hydraulic behavior with rigid multibody mechanics. The system architecture is valve-controlled, comprising a fixed-displacement pump, a gas-charged accumulator, and 4/3 proportional directional control valves (DCVs) driving double-acting cylinders. Figure 2 shows the electro-hydraulic actuator subsystem.

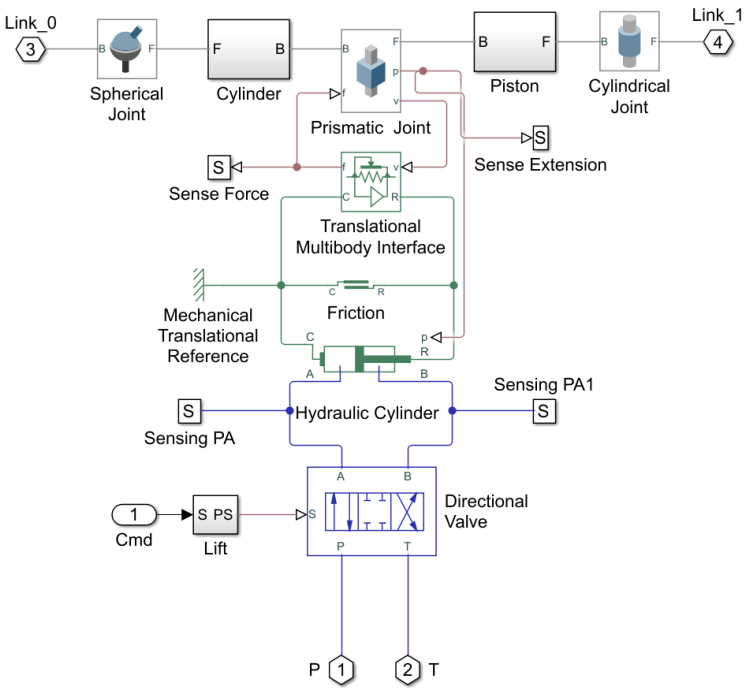


Fig. 2: Simscape implementation of the electro-hydraulic actuator subsystem. The model features a tight coupling between the physical domains: the Hydraulic Cylinder block drives the mechanical Prismatic Joint via a translational interface. Integrated sensors provide position and pressure feedback (p_A, p_B) to the control layer, while the 4/3 directional valve regulates flow based on the command input.

A. Fluid Properties and Supply Dynamics

The dynamic response of the hydraulic system is governed by the compressibility of the working fluid. This model accounts for the effective bulk modulus E_{eff} to capture the stiffness reduction caused by entrained air, particularly at

lower operating pressures [2]. The effective modulus is defined as:

$$E_{\text{eff}}(p) = \left(\frac{V_{\text{oil}}}{E_{\text{oil}}} + \frac{V_{\text{gas}}}{E_{\text{gas}}} \right)^{-1} \approx \left(\frac{1}{E_f} + \frac{\alpha}{p} \right)^{-1}, \quad (15)$$

where E_f is the fluid modulus and α represents the entrained air volume fraction. To ensure consistent pressure supply and reject flow disturbances, the circuit uses a gas-charged accumulator modeled via polytropic gas dynamics ($pV^n = \text{const}$).It is critical for smoothing the pressure ripple from the fixed-displacement pump ($Q_p = D_p\omega$) and improving the energy efficiency of the system during high-transient maneuvers [13]. Figure 5 in the appendix section shows the integrated multi-domain blocks of the Mulag FME500.

B. Valve-Controlled Actuator Dynamics

The core actuation is provided by asymmetric double-acting cylinders. A critical contribution of this model is the explicit handling of the differential area dynamics and chamber dead volumes, which are often neglected in linearized models but are essential for accurate force tracking in robotic manipulation [4].

The pressure dynamics in the cap-side (p_1) and rod-side (p_2) chambers are derived from the continuity equation:

$$\begin{aligned} \dot{p}_1 &= \frac{E_{\text{eff}}}{V_1(x)} \left(Q_1 - A_1\dot{x} - C_{\text{leak}}(p_1 - p_2) \right), \\ \dot{p}_2 &= \frac{E_{\text{eff}}}{V_2(x)} \left(-Q_2 + A_2\dot{x} + C_{\text{leak}}(p_1 - p_2) \right), \end{aligned} \quad (16)$$

where A_1, A_2 are the piston areas, x is the actuator stroke, and C_{leak} represents the internal laminar leakage coefficient. The chamber volumes $V_{1,2}(x)$ incorporate both the stroke-dependent volume and the connecting line dead volumes, which significantly influence the hydraulic natural frequency.

1) Proportional Valve Flow: Flow into the chambers (Q_1, Q_2) is regulated by a 4/3 proportional valve. To capture the physical bandwidth limitations of the solenoid, the valve spool position x_v is modeled as a first-order dynamic system rather than an instantaneous gain:

$$\dot{x}_v = \frac{1}{\tau_v}(u_{\text{cmd}} - x_v), \quad (17)$$

where τ_v is the valve time constant and u_{cmd} is the normalized control input. The resulting flow follows the nonlinear orifice equation [2]:

$$Q_j = C_d\, w\, x_v \sqrt{\frac{2}{\rho}\Delta p_j}, \quad j \in \{1, 2\}, \quad (18)$$

where C_d is the discharge coefficient and w is the valve area gradient.

C. Actuator Force and Friction Modeling

The hydraulic force generated by the actuator is given by the differential pressure acting across the piston areas, opposed by significant nonlinear friction:

$$F_h = p_1 A_1 - p_2 A_2 - F_{\text{fric}}(\dot{x}). \quad (19)$$

To improve upon standard viscous-only friction models, this work implements a parameterized friction model (Stribeck effect) to capture the stick-slip phenomena observed at low velocities [14]. This is crucial for the precise positioning required by the Mulag arm:

$$F_{\text{fric}}(\dot{x}) = \sigma_v\dot{x} + \text{sgn}(\dot{x}) \left[F_C + (F_S - F_C)e^{-|\dot{x}/v_s|^\delta} \right], \quad (20)$$

where F_C is Coulomb friction, F_S is static friction, and v_s is the Stribeck velocity.

D. Multibody Interface and Implementation

The derived hydraulic forces are coupled to the mechanical structure via a rigid multibody interface. Unlike decoupled simulations, this study integrates the hydraulic state equations directly with the manipulator's equations of motion in a unified solver environment. The actuator stroke x is mapped to the joint space configuration $\mathbf{q}$ via the kinematic Jacobian $J(\mathbf{q})$:

$$\tau_{\text{joint}} = J(\mathbf{q})^T F_h. \quad (21)$$

The complete system is implemented in Simscape Fluids/Multibody, allowing for the evaluation of a control strategy against realistic physical constraints and variable stiffness.

IV. Control-Based Validation and Experimental Results

The fidelity of the virtual prototype is validated using a closed-loop trajectory tracking task. A standard joint-space controller serves as a fixed excitation source. This approach ensures that variations in performance metricsspecifically actuation delays and tracking errors are attributable solely to the modeled physical parameters rather than control adaptation.

A. Integral Backstepping Formulation

A decentralized integral backstepping controller is implemented on all five axes with a fixed sampling time of $T_s = 0.01$ s. The control design stabilizes the error dynamics by defining a virtual velocity reference.

For each joint i, the position tracking error is defined as $z_1 = q_{ref} - q_{meas}$. A stabilizing function α is computed to drive z_1 to zero, introducing a velocity error variable z_2:

$$\begin{aligned} \alpha &= \dot{q}_{ref} + c_1 z_1 \\ z_2 &= \alpha - \dot{q}_{meas} \end{aligned} \quad (22)$$

where $c_1 > 0$ is a design gain. To reject steady-state disturbances caused by gravity and unmodeled friction, an integral action Z_{int} accumulates the velocity error z_2.

The final control law includes a model-based feedforward term for friction compensation (U_{fric}) and is normalized by the system gain estimate b_{sys}. The implementation details, including discrete-time integration and signal saturation to the valve range $u \in [-1, 1]$, are summarized in Algorithm 1.

B. Performance Metrics

Two complementary classes of metrics are used to assess the fidelity of the virtual prototype: the *temporal response* of the hydraulic actuation and the *spatial accuracy* of the end-effector trajectory.

Algorithm 1 Integral Backstepping Control Algorithm

Require: Reference $\mathbf{q}_{ref}, \dot{\mathbf{q}}_{ref}$, Measurement $\mathbf{q}_{meas}, \dot{\mathbf{q}}_{meas}$
Ensure: Valve command $\mathbf{u} \in \mathbb{R}^5$
Parameters: $\mathbf{c}_1, \mathbf{c}_2, \mathbf{K}_{int}, \mathbf{U}_{fric}, \mathbf{b}_{sys}$
for each joint $i = 1$ to 5 **do**
  $z_1 \leftarrow q_{ref}(i) - q_{meas}(i)$
  $\alpha \leftarrow \dot{q}_{ref}(i) + c_1(i) z_1$
  $z_2 \leftarrow \alpha - \dot{q}_{meas}(i)$
  $Z_{int}(i) \leftarrow Z_{int}(i) + z_2 \cdot T_s$ ▷ Discrete Integration
  Compute Control Law:
  $v \leftarrow z_1 + c_2(i) z_2 + K_{int}(i) Z_{int}(i)$
  $u_{raw} \leftarrow (v + U_{fric}(i)\mathrm{sgn}(z_2))/b_{sys}(i)$
  $u(i) \leftarrow \max(-1, \min(1, u_{raw}))$ ▷ Saturate
end for
return $\mathbf{u}$

1) Hydraulic Actuation Timing: For each hydraulic joint $j \in \{1, \dots, 4\}$ the following signals are recorded at a sampling time $T_s = 0.01$ s: the valve command $u_j(t)$, the cylinder stroke $x_j(t)$, the differential pressure $\Delta p_j(t) = p_{A,j}(t) - p_{B,j}(t)$, and the joint position $q_j(t)$. From these time histories four characteristic instants are extracted:

- $T_0^{(j)}$: *command onset*, defined as the first sample at which $|u_j(t) - u_j(t_0)|$ exceeds a small threshold.
- $T_1^{(j)}$: time at which the cylinder stroke deviation $|x_j(t) - x_j(t_0)|$ reaches 50% of its maximum value over the manoeuvre.
- $T_2^{(j)}$: time at which the pressure difference deviation $|\Delta p_j(t) - \Delta p_j(t_0)|$ reaches 50% of its maximum.
- $T_3^{(j)}$: time at which the joint angle deviation $|q_j(t) - q_j(t_0)|$ reaches 50% of its maximum.

The times $T_k^{(j)}$ are obtained with sub-sample resolution by linear interpolation between consecutive samples in which the respective thresholds are crossed. On this basis several delay measures are defined:

$$\tau_{\mathrm{v}\to\mathrm{cyl}}^{(j)} = T_1^{(j)} - T_0^{(j)}, \tag{23}$$

$$\tau_{\mathrm{v}\to\mathrm{press}}^{(j)} = T_2^{(j)} - T_0^{(j)}, \tag{24}$$

$$\tau_{\mathrm{v}\to\mathrm{joint}}^{(j)} = T_3^{(j)} - T_0^{(j)}, \tag{25}$$

$$\tau_{\mathrm{press}\to\mathrm{cyl}}^{(j)} = T_1^{(j)} - T_2^{(j)}, \tag{26}$$

$$\tau_{\mathrm{press}\to\mathrm{joint}}^{(j)} = T_3^{(j)} - T_2^{(j)}. \tag{27}$$

The primary quantity of interest is the macroscopic *valve-to-joint delay* $\tau_{\mathrm{v}\to\mathrm{joint}}^{(j)}$, which captures the combined lag of valve dynamics, fluid compressibility and mechanical transmission. For a compact system-level indicator, a *Delay Index* is defined as the root-mean-square (RMS) of the four valve-to-joint delays:

$$\tau_{\mathrm{RMS}} = \sqrt{\frac{1}{4}\sum_{j=1}^{4}\left(\tau_{\mathrm{v}\to\mathrm{joint}}^{(j)}\right)^2}. \tag{28}$$

2) Task-Space Tracking Error: Spatial performance is quantified at the mower head (end-effector). Let $p_{\mathrm{ref}}(t) \in \mathbb{R}^3$ denote the reference trajectory and $p_{\mathrm{meas}}(t)$ the simulated end-effector position. The instantaneous task-space error is

$$e(t) = \left\| p_{\mathrm{ref}}(t) - p_{\mathrm{meas}}(t) \right\|_2. \tag{29}$$

Over the active manoeuvre window $[t_{\mathrm{start}}, t_{\mathrm{end}}]$ the global performance is summarized by the RMS and mean absolute error:

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{T}\int_{t_{\mathrm{start}}}^{t_{\mathrm{end}}} e^2(t)\,\mathrm{d}t}, \tag{30}$$

$$E_{\mathrm{mean}} = \frac{1}{T}\int_{t_{\mathrm{start}}}^{t_{\mathrm{end}}} |e(t)|\,\mathrm{d}t, \tag{31}$$

with $T = t_{\mathrm{end}} - t_{\mathrm{start}}$. These metrics are reported in millimetres in Table II.

V. Results and Discussion

The validated model is used to investigate the sensitivity of the system to the effective hydraulic bulk modulus β. Four scenarios are simulated, with $\beta \in \{0.5, 1.0, 1.5, 2.0\}$ GPa. For ease of reference we will refer informally to these as "soft", "nominal", "stiff" and "very stiff" fluid, while all controller gains are kept constant.

A. Baseline Tracking Validation

Before analyzing the parametric sensitivity, the baseline fidelity of the plant is verified. Figure 3 illustrates the task-space trajectory tracking under nominal conditions ($\beta = 1.0$ GPa). The end-effector path closely follows the reference mowing contour, and no instability or drift is observed over the full manoeuvre. This confirms that the coupled electro-hydraulic model, including the controller, is suitable for subsequent sensitivity analysis.

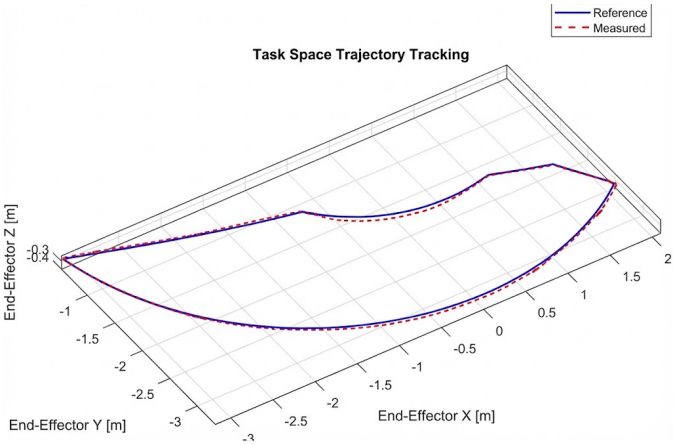


Fig. 3: Task-space trajectory tracking validation for the nominal bulk modulus ($\beta = 1.0$ GPa). The simulated end-effector path closely follows the reference trajectory.

B. Influence on Actuation Delay

Table I reports the valve-to-joint 50% delays for the dominant axes (Joints 1, 3 and 4) and the corresponding Delay Index τ_{RMS}. Joint 2 is omitted from the table because its delay

remains below 0.3 ms for all values of β, i.e., well below the time scales of interest in this manoeuvre.

TABLE I: Valve-to-joint 50% delays and delay index for different bulk moduli.

β [GPa]	$\tau^{(1)}_{v\to\text{joint}}$ [ms]	$\tau^{(3)}_{v\to\text{joint}}$ [ms]	$\tau^{(4)}_{v\to\text{joint}}$ [ms]	τ_{RMS} [ms]
0.5	847.4	186.3	202.1	445.4
1.0	725.3	168.8	206.4	386.4
1.5	674.0	149.9	215.8	361.7
2.0	649.7	152.4	221.6	351.6

Joint 1 exhibits the largest macroscopic delay, with $\tau^{(1)}_{\text{v}\to\text{joint}}$ ranging from approximately 0.85 s at $\beta = 0.5$ GPa to 0.65 s at $\beta = 2.0$ GPa. This axis therefore dominates the overall actuation speed of the arm. Joints 3 and 4 show intermediate delays between 0.15 and 0.22 s, consistent with their lighter effective inertia and more favorable leverage. Their dependence on β is weaker but still follows the expected trend of decreasing delay with increasing stiffness, followed by saturation. Joint 2 displays a negligible delay for this manoeuvre: the 50% motion threshold is reached within the same sample as the command onset for all values of β, and $\tau^{(2)}_{\text{v}\to\text{joint}}$ remains below 0.3 ms. This should not be interpreted as a truly delay-free axis, but rather as an indication that its closed-loop response is much faster than the other joints and below the resolution of the chosen metric.

Aggregating the four joints into the Delay Index τ_{RMS} highlights the impact of fluid compressibility at the system level. As β increases from 0.5 to 2.0 GPa, the index decreases from approximately 0.45 s to 0.35 s, corresponding to a reduction of about 20% in macroscopic actuation delay. The improvement is most pronounced when moving from the soft fluid case (0.5 GPa) to the nominal range (1.0–1.5 GPa); further stiffening beyond 1.5 GPa yields only marginal gains, indicating that the response becomes dominated by rigid-body inertia and controller tuning rather than fluid compressibility.

The detailed timing also allows a more focused analysis of the heavy boom axis (Joint 3). The corresponding delay $\tau^{(3)}_{\text{v}\to\text{joint}}$ (reported again in Table II) decreases from roughly 185 ms at $\beta = 0.5$ GPa to 155 ms at $\beta = 1.5$ GPa and slightly increases again for $\beta = 2.0$ GPa. This weakly convex behaviour suggests that, beyond a certain stiffness, the local dynamics are limited by the mechanical configuration and actuator flow capacity rather than by fluid compressibility alone.

It is worth noting that the negligible delay observed for Joint 2 is still consistent with industrial specifications for modern proportional and servo valves, which typically exhibit step-response times on the order of 5–50 ms [15]–[17]. In the present manoeuvre, the required motion of Joint 2 is small and its effective closed-loop bandwidth is high, such that its contribution to the global delay is negligible compared to the dominant axes.

C. Influence on Tracking Accuracy

Although the actuation delays of the individual joints show a non-uniform dependence on β, the global spatial accuracy exhibits a clear and monotonic trend. The RMS task-space error E_{RMS} and the mean absolute error E_{mean} are summarized in Table II. As the bulk modulus increases from 0.5 to 2.0 GPa, E_{RMS} improves from 0.83 m to 0.74 m, corresponding to an improvement of approximately 11%. The mean error follows a similar pattern.

TABLE II: Sensitivity of Joint Delays and End-Effector Accuracy to Bulk Modulus.

Case	β [GPa]	$\tau^{(3)}$ [ms]	E_{RMS} [mm]	E_{mean} [mm]
Soft Fluid	0.5	185.0	827.9	211.3
Nominal	1.0	166.5	775.6	215.4
Stiff Fluid	1.5	155.3	753.2	229.1
Very Stiff Fluid	2.0	173.7	739.4	228.8

This behaviour confirms that a stiffer hydraulic transmission enhances the positional stiffness of the arm and reduces compliance-induced deflection at the mower head. In other words, while the Delay Index saturates for large β, the end-effector still benefits from increased fluid stiffness through improved spatial precision. Figure 4 in the appendix section visualizes this decoupling: the end-effector tracking error decreases monotonically with β, whereas the boom-joint delay $\tau^{(3)}_{\text{v}\to\text{joint}}$ exhibits only a mild convex dependence and the other joints show even weaker sensitivity.

VI. Conclusion and Future Work

This work presented a high-fidelity dynamic model of the Mulag FME500 hydraulic manipulator, bridging the gap between rigid multibody mechanics and nonlinear hydraulic dynamics. By moving beyond simplified kinematic assumptions, the proposed simulation environment explicitly captures the bidirectional coupling between actuator forces and structural loads.

The control-based validation demonstrated that the virtual prototype exhibits realistic physical behaviors. The results confirmed that the model accurately captures the sensitivity of actuation delays to fluid compressibility. Specifically, the observed increase in delay with reduced bulk modulus verifies the model's ability to represent the "stiff" yet oscillatory nature of hydraulic machinery, providing a rigorous foundation for virtual prototyping.

Future research will leverage this high-fidelity simulation as a digital twin for data-driven control methods. Specifically, the focus will shift to learning the full system dynamics using Reinforcement Learning (RL) techniques to identify the complex, nonlinear hydraulic-mechanical coupling. These learned models will subsequently support the development of advanced control architectures capable of compensating for actuation delays to achieve high-precision tracking.

Appendix

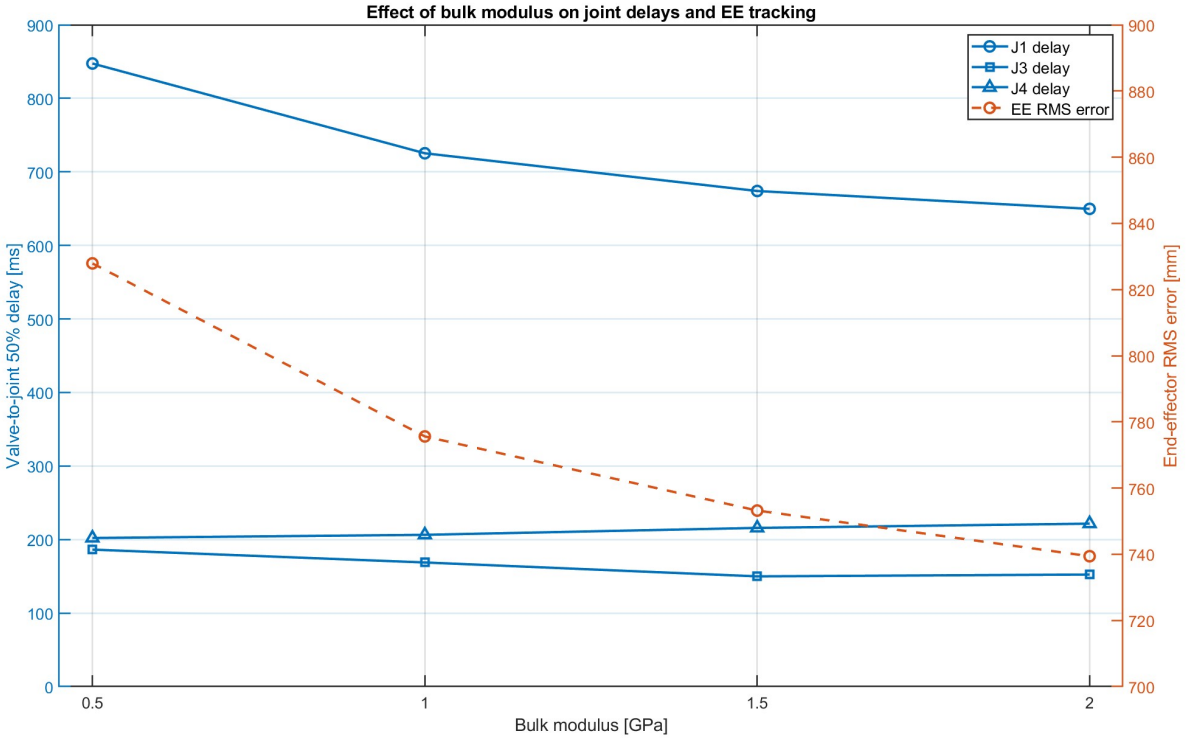


Fig. 4: Effect of bulk modulus on joint delays and task-space tracking. The left axis shows the valve-to-joint 50% delays for Joints 1, 3 and 4, while the right axis shows the end-effector RMS tracking error E_{RMS}.

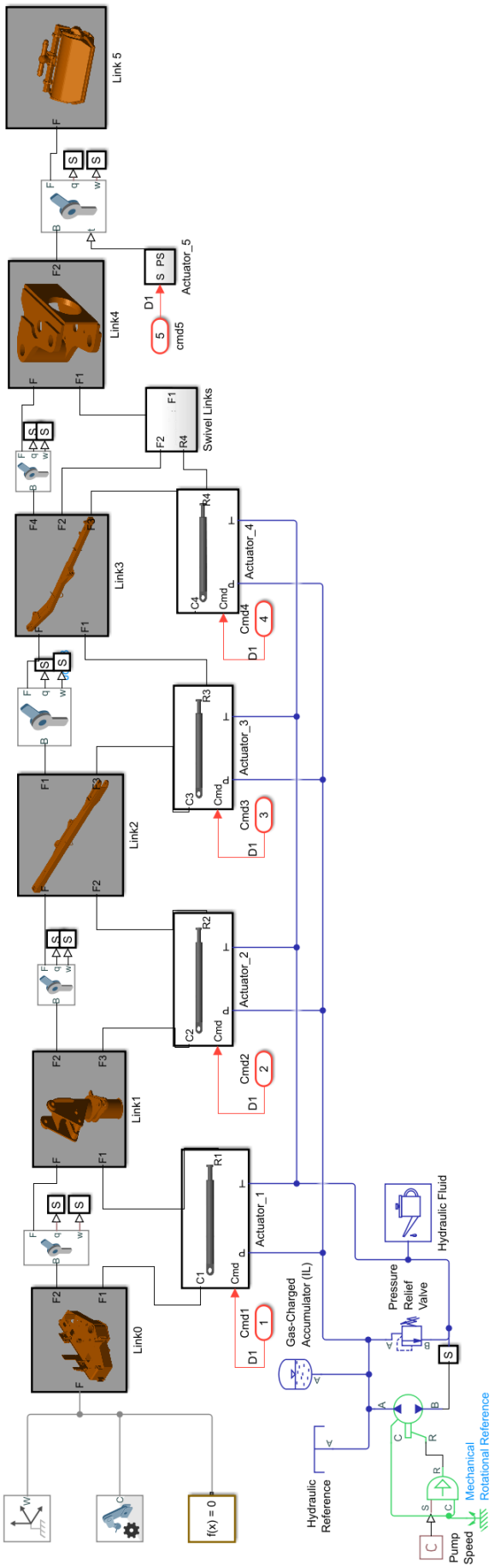


Fig. 5: Integrated multi-domain simulation environment of the Mulag FME 500. The architecture couples the rigid multibody mechanism (Links 0–5) directly with the hydraulic power network. Key subsystems include the variable-speed pump source, the gas-charged accumulator for pressure stabilization, and the five distributed electro-hydraulic actuators.

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